Model	Arch.	Layer	Acc.
S4	No gate	S4	18.3
-	No gate	S6	**97.0**
H3	H3	S4	57.0
Hyena	H3	Hyena	30.1
-	H3	S6	**99.7**
-	Mamba	S4	56.4
-	Mamba	Hyena	28.4
Mamba	Mamba	S6	**99.8**

Table 1: (**Selective Copying**.) Accuracy for combinations of architectures and inner sequence layers.

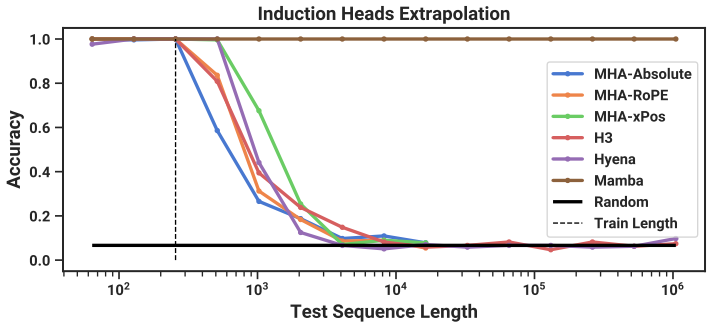


Table 2: (**Induction Heads**.) Models are trained on sequence length $2^8 = 256$, and tested on increasing sequence lengths of $2^6 = 64$ up to $2^{20} = 1048576$. Full numbers in Table 11.

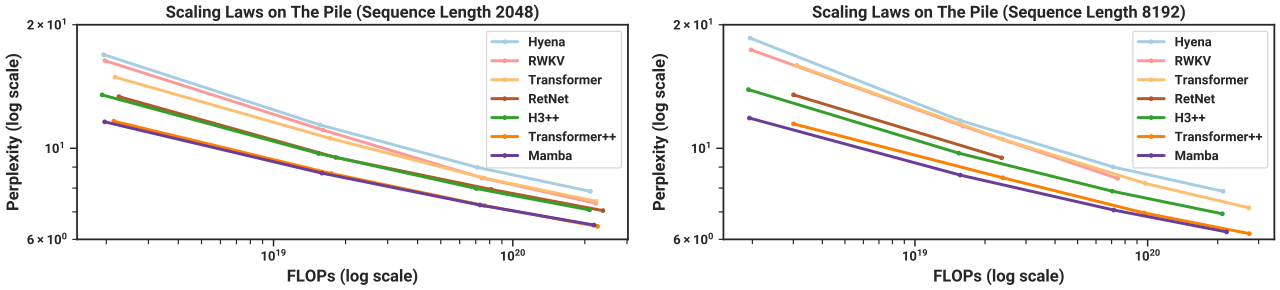


Figure 4: (**Scaling Laws**.) Models of size $\approx 125M$ to $\approx 1.3B$ parameters, trained on the Pile. Mamba scales better than all other attention-free models and is the first to match the performance of a very strong "Transformer++" recipe that has now become standard, particularly as the sequence length grows.

Out of positional encoding variants for attention models, xPos (which was designed for length extrapolation) is slightly better than the others; also note that all attention models were only tested up to sequence length $2^{14} = 16384$ due to memory limitations. Out of other SSMs, H3 and Hyena are similar, contrary to the findings in Poli et al. (2023).

4.2 Language Modeling

We evaluate the Mamba architecture on standard autoregressive language modeling against other architectures, on both pretraining metrics (perplexity) and zero-shot evaluations. We set the model sizes (depth and width) to mirror GPT3 specifications. We use the Pile dataset (L. Gao, Biderman, et al. 2020), and follow the training recipe described in Brown et al. (2020). All training details are in Appendix E.2.

4.2.1 Scaling Laws

For baselines, we compare against the standard Transformer architecture (GPT3 architecture), as well as the strongest Transformer recipe we know of (here referred to as Transformer++), based on the PaLM and LLaMa architectures (e.g. rotary embedding, SwiGLU MLP, RMSNorm instead of LayerNorm, no linear bias, and higher learning rates). We also compare against other recent subquadratic architectures (Figure 4). All model details are in Appendix E.2.

Figure 4 shows scaling laws under the standard Chinchilla (Hoffmann et al. 2022) protocol, on models from $\approx 125M$ to $\approx 1.3B$ parameters. **Mamba is the first attention-free model to match the performance of a very strong Transformer recipe (Transformer++) that has now become standard, particularly as the sequence length grows.** (We note that full results on context length 8k are missing for the RWKV and RetNet baselines, prior strong recurrent models that can also be interpreted as SSMs, because of a lack of efficient implementations leading to out-of-memory or unrealistic computation requirements.)

Table 13: (**Great Apes DNA Classification**.) Accuracy after fine-tuning on sequences of length $2^{10} = 1024$ up to $2^{20} = 1048576$ using pretrained models of the same context length. Random guessing is 20%.

Model	Params	Accuracy (%) at Sequence Length					
		2^{10}	2^{12}	2^{14}	2^{16}	2^{18}	2^{20}
HyenaDNA	1.4M	28.04	28.43	41.17	42.22	31.10	54.87
Mamba	1.4M	31.47	27.50	27.66	40.72	42.41	**71.67**
Mamba	7M	30.00	29.01	31.48	43.73	56.60	**81.31**

Table 14: YouTubeMix length scaling sequence lengths and batch sizes.

Sequence length	Batch size	Tokens / batch
$468 \times 2048 = 958464$	1	958464
$234 \times 2048 = 479232$	2	958464
$117 \times 2048 = 239616$	4	958464
$59 \times 2048 = 120832$	8	966656
$30 \times 2048 = 61440$	16	983040
$15 \times 2048 = 30720$	32	983040
$8 \times 2048 = 16384$	64	1048576
$4 \times 2048 = 8192$	128	1048576

warmup was also long (e.g. comprising 6 out of 10 epochs for the model with context length 2^{20}); we did not experiment with this choice.

Results for the Species classification task are in Table 13.

E.4 Audio Details

E.4.1 YouTubeMix Audio Pretraining

Model. We use a model with 3 blocks per stage ($3 \times 5 = 15$ total Mamba blocks), pooling factor $p = 16$, and outer dimension $D = 64$, for about 3.5M parameters.

Dataset. The data is mu-law encoded at 8 bits, so the model is modeling discrete tokens with a vocab size of 256.

The dataset consists of clips of up to 1 minute long, or length 960000, which is subsampled and divided into segments of any desired sequence length. Since the architecture involves two stages of pooling by a factor of 16, and we want the resulting sequence length to be a a multiple of 8 for hardware efficiency, the longest possible sequence is $468 \times 2048 = 958464$. The rest of our sequence lengths are defined by successively halving this and rounding up to the nearest multiple of 2048.

Table 14 lists the specifications used in Figure 7. Beyond the varying batch sizes, the number of valid segments in the training set varied between different sequence lengths (e.g. the number of training steps per epoch was not constant for different points in the graph), which may have contributed to kinks in the scaling curves.

Training. Models were trained for $200K$ training steps with a maximum learning rate of 0.002, $20K$ (10%) warmup steps, and weight decay 0.1 (similar to our general pretraining recipe across domains).

Additional Ablations: SSM Parameterizations. We investigate SSM parameterizations on long-form audio waveform pretraining in the setting of Figure 7. The setting is modified slightly to use larger models (8 layers and $D = 64$ for 6M params, the SaShiMi default), shorter sequences ($2^{11} = 2048$ to $2^{18} = 262144$ instead of 2^{13} to 2^{20}), lower LR (0.001 from 0.002), and shorter training cycles (100K instead of 200K steps).

Figure 10 shows that the change from S4 $\rightarrow$ S6 (i.e. the selection mechanism) is not always beneficial. On long-form audio waveforms, it in fact significantly hampers performance, which may be intuitive from the point of view that audio